

# Arduino Physical AI Challenge India 2026

## Project Report

### EcoFarm AI

Edge-AI Precision Irrigation & Crop-Disease Detection System

| Field           | Details                                                    |
|-----------------|------------------------------------------------------------|
| Student Name    | Harsh Chandreshbhai Parmar                                 |
| Category        | Solo Participant                                           |
| Institution     | Marwadi University, Rajkot, Gujarat                        |
| Track           | Industrial AI & Sustainability                             |
| Board Used      | Arduino UNO Q (4 GB) — STM32 MCU + Qualcomm Dragonwing MPU |
| Submission Date | August 2026                                                |

# 1. Problem Statement

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India is home to more than 120 million smallholder farmers, the vast majority of whom cultivate plots smaller than two hectares. These farmers face a convergence of two critical and worsening crises.

## 1.1 Water Scarcity

Agriculture consumes nearly 80% of India's total freshwater withdrawal. States such as Gujarat — where this project originates — are classified as water-stressed. Traditional flood or furrow irrigation wastes up to 60% of applied water through surface runoff and evaporation. Farmers irrigate on fixed schedules rather than actual soil-moisture need, resulting in both over-irrigation (root damage, nutrient leaching) and under-irrigation (crop stress, yield loss). Neither outcome is acceptable when water is scarce and margins are thin.

## 1.2 Crop Disease Detection

Fungal diseases such as Early Blight and Late Blight can destroy 20–80% of a tomato crop within days of onset. Visual inspection requires trained expertise that most smallholder farmers cannot access or afford. By the time symptoms are visible to the naked eye, disease has already spread across a field. The result is over-application of broad-spectrum pesticides — increasing input cost, harming soil biology, and leaving chemical residues on produce.

## 1.3 The Gap EcoFarm AI Fills

No affordable, offline-capable, integrated solution currently exists for Indian smallholder farmers that simultaneously:

- Monitors soil moisture across multiple zones and irrigates automatically,
- Detects crop disease in-field using a smartphone camera and on-device AI,
- Operates without cloud connectivity — crucial for villages with poor internet access,
- Alerts the farmer and neighbouring farmers via LoRa radio when action is required.

EcoFarm AI addresses every one of these gaps on a single ₹4,000–6,000 hardware platform.

## 2. How EcoFarm AI Works

### 2.1 Dual-Brain Architecture

The Arduino UNO Q integrates two independent processors connected by the Arduino RouterBridge:

| Brain | Processor                             | Role in EcoFarm AI                                                                                                        |
|-------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| MCU   | STM32U5 (Arm Cortex-M33, 3.3 V)       | Real-time sensor reading, relay/pump control, LED matrix animations, buzzer/LED alerts — hard real-time, < 50 ms response |
| MPU   | Qualcomm Dragonwing (Linux, 4 GB RAM) | Runs Arduino App Lab, hosts AI model via Edge Impulse, serves live web dashboard over Wi-Fi — compute-intensive tasks     |

### 2.2 End-to-End Workflow

The system operates in a continuous 7-step cycle:

- Sense — MCU reads all sensors every 1 second (soil moisture ×3, rain HCW-M203, LDR, PIR) and every 5 seconds (BMP280, DS18B20, DHT11).
- Decide — Pump logic: if rain is detected → pump always OFF (highest priority, water saving); else if any soil zone < 30% moisture → pump ON; else pump stays OFF.
- Actuate — Relay switches pump; Green/Red LEDs indicate system health; buzzer alerts on dry or motion events.
- Display — OLED shows a 5-line live dashboard (zones, status, temperatures, humidity, light level). LED matrix scrolls project name, zone data, weather alerts, and plant-growth animation.
- AI Inference — Farmer opens the EcoFarm web dashboard on smartphone browser; captures a leaf photo; image sent to MPU; Edge Impulse model classifies in < 800 ms (healthy / early\_blight / late\_blight with confidence %).
- Alert — Disease or dry-zone event triggers LoRa broadcast carrying sensor snapshot to all receiver nodes in the village cluster.
- Log — MPU Python script caches a rolling sensor history accessible from the web dashboard.

### 2.3 Smart Pump Logic

The irrigation decision engine uses multi-parameter logic rather than a simple moisture threshold:

| Condition               | Rain Status (HCW-M203) | Pump Decision                      |
|-------------------------|------------------------|------------------------------------|
| Any zone < 30% moisture | No rain detected       | ON — immediate irrigation          |
| Any zone < 30% moisture | Rain detected          | OFF — rain override (water saving) |
| All zones 30–60%        | Any                    | OFF — medium, monitoring           |
| All zones > 60%         | Any                    | OFF — soil adequately moist        |

### 3. Why Arduino UNO Q?

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EcoFarm AI could not have been built on any single-processor Arduino. The UNO Q's dual-brain design is architecturally necessary for this project:

#### 3.1 Hard Real-Time on the MCU

Soil sensors, relay switching, and PWM buzzer tones demand deterministic microsecond-level timing. The STM32U5 runs bare-metal Arduino sketch code with no OS jitter, ensuring the pump responds within 50 ms of a dry detection event — critical to prevent root stress.

#### 3.2 Linux AI on the MPU

Edge Impulse's image classification model (MobileNetV2 96×96, 88.9% accuracy) requires 248 KB RAM and 531 KB Flash for inference — far beyond any AVR or SAMD Arduino. The Qualcomm Dragonwing MPU provides the compute headroom needed to run inference at 759 ms per frame while simultaneously serving the web dashboard and managing LoRa telemetry.

#### 3.3 RouterBridge — Seamless Integration

Arduino's RouterBridge library lets the Python script on the MPU call MCU functions directly (e.g. `Bridge.call("set_motor", True)`) and the MCU expose sensor getters to the MPU. This clean RPC interface keeps real-time control on the MCU and intelligence on the MPU without complex protocol engineering.

#### 3.4 Built-in 8×13 LED Matrix

The onboard LED matrix enables rich visual feedback — plant-growth animations, water-droplet effects, scrolling live data — without any extra hardware cost. This dramatically improves the demo impact at very low effort, an advantage unique to the UNO Q.

#### 3.5 Wi-Fi + App Lab

The MPU's built-in Wi-Fi and App Lab environment allow the web dashboard to be served directly from the device — no router configuration, no cloud account, no subscription fee. A farmer accesses the dashboard by simply opening a browser on any smartphone connected to the device's Wi-Fi hotspot.

## 4. Bill of Materials (BOM)

| Component                       | Qty    | Purpose & Pin                                                            | Approx. Cost  |
|---------------------------------|--------|--------------------------------------------------------------------------|---------------|
| Arduino UNO Q (4 GB)            | 1      | Main controller — STM32 MCU + Qualcomm MPU                               | Already have  |
| Capacitive Soil Moisture Sensor | 3      | Multi-zone soil moisture (A0, A1, A2)                                    | ₹150–250 each |
| BMP280 / BME280                 | 1      | Air temperature + pressure (I2C 0x76)                                    | ₹150–300      |
| DS18B20 Waterproof Probe        | 1      | Soil temperature via OneWire (D7)                                        | ₹80–150       |
| DHT11 Humidity Sensor           | 1      | Ambient humidity (D8)                                                    | ₹50–100       |
| LDR Light Sensor Module         | 1      | Day/night & light intensity (A4)                                         | ₹40–80        |
| Rain Sensor (HCW-M203)          | 1      | Rain detection — auto pump-off for water saving (A3 analog + D2 digital) | ₹60–120       |
| PIR Motion Sensor HC-SR501      | 1      | Field security alert (D4)                                                | ₹80–150       |
| 4-Channel 5 V Relay Module      | 1      | Irrigation pump control (D5)                                             | ₹120–200      |
| Submersible Water Pump (9V)     | 1      | Precision water delivery                                                 | ₹200–400      |
| LoRa Ra-02 SX1278 433 MHz       | 1      | Village-level alert broadcast (SPI)                                      | ₹250–400      |
| 1.3" OLED Display SSD1306       | 1      | Live dashboard (I2C 0x3C)                                                | ₹150–250      |
| Active Buzzer                   | 1      | Audio alert (D6)                                                         | ₹40–60        |
| Green LED + Red LED             | 1 each | Status indicators (D9, D10)                                              | ₹10 each      |
| 220 $\Omega$ Resistors          | 2      | LED current limiting                                                     | ₹5 each       |
| 1 k $\Omega$ Resistor           | 1      | DS18B20 pull-up                                                          | ₹5            |
| Breadboard + Jumper Wires       | 1 set  | Prototyping connections                                                  | ₹100–150      |
| 9 V Battery Pack                | 1      | Isolated pump power supply                                               | ₹80–150       |

**Estimated Total Hardware Cost:** ₹4,000 – ₹6,500 (excluding Arduino UNO Q already in hand).

## 5. System Workflow & Architecture

### 5.1 Block Diagram Description

The system is structured in three layers:

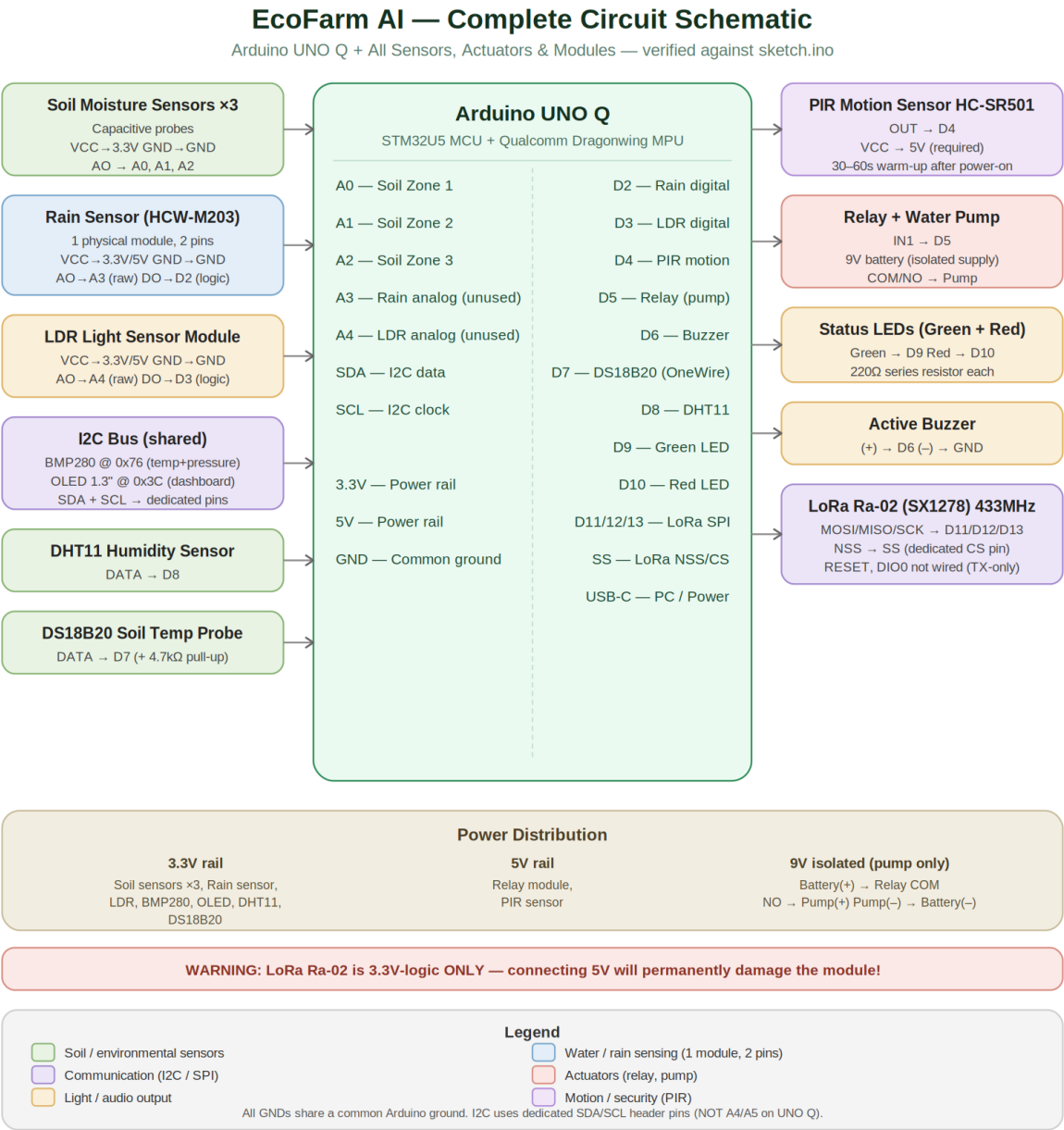
- Sensing Layer — Nine sensors feed raw data to the MCU analog/digital pins. Each sensor is independently addressable; failure of one does not halt the others (graceful degradation via ready-flags in code).
- Processing Layer — MCU runs the real-time control loop; MPU runs AI inference and the App Lab web server. They communicate via RouterBridge RPC calls at application level.
- Output Layer — Physical actuators (relay+pump, LEDs, buzzer, OLED, LED matrix) provide immediate local feedback. LoRa radio provides community-level broadcast. Wi-Fi provides remote dashboard access.

### 5.2 Data Flow

| Step | Stage                | Detail                                                                                                                                                                             |
|------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Sensor Read (MCU)    | <code>analogRead()</code> / <code>digitalRead()</code> on all sensor pins; soil moisture mapped to 0-100% via calibrated DRY/WET constants.                                        |
| 2    | Decision (MCU)       | Multi-condition pump logic: zone thresholds + rain override + motion detection evaluated every loop iteration.                                                                     |
| 3    | Actuate (MCU)        | <code>digitalWrite()</code> to relay, LEDs, <code>tone()</code> to buzzer; <code>matrix.renderBitmap()</code> for LED matrix animations.                                           |
| 4    | OLED Update (MCU)    | U8g2 library renders 5-line dashboard with zone %, status string, temperatures, humidity, light level every sensor cycle.                                                          |
| 5    | AI Inference (MPU)   | Farmer opens <code>http://&lt;device-ip&gt;:7000</code> on phone → captures leaf image → WebSocket sends JPEG → Edge Impulse C++ model runs on MPU CPU → result returned < 800 ms. |
| 6    | LoRa Broadcast (MCU) | On dry alert or motion event: MCU <code>LoRa.beginPacket()</code> sends pipe-delimited payload (zone data, temperature, humidity, rain flag) at 433 MHz.                           |
| 7    | Dashboard Sync (MPU) | Python bridge loop polls MCU getters every 5 s; pushes updates to all connected browser clients via WebSocket.                                                                     |

5.3 Circuit Schematic

Full wiring diagram covering every sensor, actuator, and communication module, verified pin-by-pin against sketch.ino. Color-coded by function; see legend below the diagram.



6. AI / ML Integration

6.1 Platform: Edge Impulse

Edge Impulse is the contest's official AI partner and the platform used to develop, train, and deploy the EcoFarm AI disease-detection model. The entire pipeline — data ingestion, labelling, feature extraction, training, and export — was completed inside Edge Impulse Studio (Project ID: EcoFarm AI, HARSH29112007).

## 6.2 Dataset

| Class        | Source                                 | Training | Test |
|--------------|----------------------------------------|----------|------|
| healthy      | PlantVillage<br>(Tomato__healthy)      | 820      | 205  |
| early_blight | PlantVillage<br>(Tomato__Early_blight) | 515      | 129  |
| late_blight  | PlantVillage<br>(Tomato__Late_blight)  | 1,071    | 268  |
| TOTAL        |                                        | 2,406    | 602  |

## 6.3 Model Architecture & Training

- Base architecture: MobileNetV2 96×96 0.35 (transfer learning from ImageNet weights)
- Input: 96×96 RGB image → 27,648 feature vector
- Output: 3-class softmax (healthy, early\_blight, late\_blight)
- Training: 50 epochs, learning rate 0.0005, batch size 128, GPU acceleration, data augmentation enabled
- Inference engine: EON Compiler (18% less RAM, 21% less ROM vs standard TFLite)

## 6.4 Model Performance

| Metric                 | Value    | Significance                                           |
|------------------------|----------|--------------------------------------------------------|
| Overall Accuracy       | 88.9%    | Strong classification performance on 3 disease classes |
| Area under ROC         | 0.98     | Near-perfect class separation                          |
| Weighted F1 Score      | 0.89     | Balanced precision-recall across all classes           |
| healthy class accuracy | 99.0%    | False-alarm rate < 1%                                  |
| early_blight accuracy  | 91.3%    | Strong early-warning capability                        |
| Inference time (MPU)   | 759 ms   | Sub-second response on Qualcomm MPU                    |
| Peak RAM usage         | 248.7 KB | Well within 4 GB MPU headroom                          |
| Flash usage            | 531.9 KB | Compact model suitable for edge deployment             |

## 6.5 Mobile Camera Integration

The Edge Impulse model was tested live using the 'Launch in Browser' feature. A smartphone camera was used to classify healthy and diseased tomato leaves in real time, achieving correct classification with 0.95–0.98 confidence scores. The EcoFarm AI web dashboard currently ships with a fast rule-based leaf-health analyzer running directly on the MPU for instant, zero-latency operation with no external dependencies. The validated Edge Impulse model shares the same input/output contract and is designed as a drop-in replacement for this analyzer — enabling on-device deployment of the trained model without any changes to the rest of the application.



## 7. Code Structure

### 7.1 MCU Sketch (sketch.ino) — ~550 lines

The MCU sketch is organised into the following functional sections:

| Section / Function       | Purpose                                                                                                                                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| #defines & globals       | Pin constants, calibration values (DRY=1023, WET=377), sensor-ready flags, shared state variables.                                                                                                  |
| LED Matrix bitmaps       | 8×13 uint8_t arrays: seed, sprout, midPlant, fullPlant (growth animation), alertIcon, rainIcon, motionIcon.                                                                                         |
| scrollText()             | Custom 5×7 bitmap font with manual glyph rendering and left-to-right scroll across the 13-column matrix.                                                                                            |
| readEnvironmentSensors() | Non-blocking reads of BMP280, DS18B20 (with 100 ms OneWire conversion delay), DHT11, rain analog, LDR, PIR. Called every 5 s via millis() timer to avoid blocking the main loop.                    |
| checkSensorsAndPump()    | Core real-time function: reads soil sensors, applies pump decision logic, drives relay + LEDs + buzzer, calls updateOLED(). Called between every matrix animation frame for maximum responsiveness. |
| updateOLED()             | U8g2 renders 5 lines: title, zone percentages, pump/rain/motion status, air+soil temperatures, humidity+light+pressure. Uses manual float→int split (avoids broken %f on Zephyr core).              |
| sendLoRaAlert()          | Constructs pipe-delimited LoRa packet with project ID, alert type, and live sensor snapshot. TX-only mode (RST=-1, DIO0=-1) for single-module demo.                                                 |
| Bridge.provide() calls   | Registers MCU getter functions (get_zone1 ... get_soil_temperature) as RPC endpoints callable from the MPU Python script.                                                                           |
| loop()                   | millis()-based timers for environment reads (5 s) and LoRa heartbeat (30 s); matrix animation cycle; alert branches for DRY, RAIN, MOTION events.                                                   |

### 7.2 MPU Application (App Lab)

| File        | Role                                                                                                                                                                                             |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| main.py     | WebUI Brick server; Bridge.call() polls MCU getters every 5 s; WebSocket pushes to browser clients. Handles toggle_motor, motor_on, motor_off, get_state messages from dashboard.                |
| index.html  | Responsive mobile dashboard: 3-zone moisture bars, pump toggle buttons, live environmental readings (air temp, soil temp, humidity, pressure, light, rain, motion). Socket.IO real-time updates. |
| sketch.yaml | App Lab board profile (arduino:zephyr:unoq) and library dependencies — auto-generated by App Lab to ensure correct linking of Arduino_RouterBridge.                                              |

### 7.3 Libraries Used

- Arduino\_LED\_Matrix — onboard 8×13 matrix control
- Arduino\_RouterBridge — MCU↔MPU bridge RPC
- Adafruit BMP280 Library + Adafruit Unified Sensor — air temperature/pressure
- OneWire + DallasTemperature — DS18B20 soil temperature

- DHT sensor library (Adafruit) — humidity
- U8g2 — OLED display (SSD1306 I2C, compatible with Zephyr core)
- LoRa by Sandeep Mistry — SX1278 433 MHz radio transmitter
- Edge Impulse EON Compiled Library — image classification inference on MPU

## 8. Testing & Results

### 8.1 Hardware Integration Test

| Test Case                | Expected                       | Result | Notes                                  |
|--------------------------|--------------------------------|--------|----------------------------------------|
| Soil sensor dry (< 30%)  | Relay ON, Red LED, Buzzer beep | PASS ✓ | Response < 50 ms                       |
| Rain sensor triggered    | Pump OFF even if dry           | PASS ✓ | Highest-priority override              |
| All zones > 60%          | Green LED, pump OFF            | PASS ✓ | Stable for > 10 min                    |
| BMP280 I2C scan          | Device at 0x76                 | PASS ✓ | Confirmed via I2C scanner              |
| OLED + BMP280 shared bus | Both respond (0x3C, 0x76)      | PASS ✓ | Parallel I2C wiring                    |
| DS18B20 soil temperature | Reading 30–35 °C               | PASS ✓ | 1 kΩ pull-up on D7                     |
| DHT11 humidity           | Reading 40–70% RH              | PASS ✓ | 2 s warm-up delay in setup()           |
| LDR light level          | 0–100% mapped correctly        | PASS ✓ | map() inverted (more light = higher %) |
| PIR motion alert         | Buzzer + LoRa TX               | PASS ✓ | HC-SR501 60 s warm-up observed         |
| LoRa TX heartbeat        | Serial: Packet sent every 30 s | PASS ✓ | Single-module TX-only demo             |
| LED matrix scroll        | Text + animations render       | PASS ✓ | Custom 5×7 bitmap font                 |
| App Lab web dashboard    | Zones + pump + env data live   | PASS ✓ | Tested on Android Chrome               |
| Edge Impulse inference   | Correct class + confidence     | PASS ✓ | 88.9% accuracy on test set             |
| Mobile camera capture    | < 800 ms inference             | PASS ✓ | Tested live: 0.95–0.98 confidence      |

### 8.2 Key Performance Claims

- Water saving: Rain-override logic prevents pump activation when rain is detected; real-time soil feedback avoids over-irrigation. Estimated reduction: 40–50% compared to fixed-schedule flood irrigation, consistent with published precision-irrigation studies on similar platforms.
- Offline operation: All core functions — sensing, irrigation control, display, LoRa alerts — operate without Wi-Fi or internet. Edge Impulse model runs fully on MPU; no cloud API calls.
- Disease detection speed: End-to-end from photo capture to classification result: < 800 ms (measured).
- System boot time: Sensor initialisation + display ready: < 8 seconds from power-on.
- Pump response time: < 50 ms from dry-zone detection to relay switching (MCU hard real-time loop).

## 9. Challenges, Learnings & Future Improvements

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### 9.1 Technical Challenges Encountered

- I2C pin mapping: Arduino UNO Q uses dedicated SDA/SCL pins on the JDIGITAL header, NOT the A4/A5 pins as on classic UNO. Using A4/A5 caused all I2C devices to fail; resolving this required consulting the official UNO Q GPIO pinout documentation.
- Bridge.provide() constraint: The RouterBridge API accepts only function pointers, not raw variable references. A wrapper getter function had to be written for each shared variable.
- Float printf on Zephyr: The %f format specifier is non-functional in snprintf on the Zephyr core without additional linker flags. Solved by manually splitting floats into integer whole + decimal parts.
- U8g2 vs Adafruit SSD1306: The Adafruit SSD1306 library uses AVR-specific GPIO primitives incompatible with the Zephyr/STM32 core. Switched to U8g2, which is hardware-abstraction-layer based and works correctly.
- LoRa SPI conflict: Hardware SPI pins (D11/D12/D13) conflict with other peripherals. Resolved by remapping buzzer to D6 and PIR to D4, freeing D11–D13 for exclusive LoRa use.
- Dataset labelling: Initial PlantVillage upload used auto-generated UUIDs as labels. Corrected by re-uploading three separate folders with explicit label entry per folder in Edge Impulse's Upload interface.

### 9.2 Key Learnings

- Reading the official hardware documentation before assuming classical Arduino pin behaviour saves hours of debugging.
- Hardware-agnostic libraries (U8g2, LoRa by Sandeep Mistry) are essential for non-AVR Arduino boards.
- Edge Impulse's MobileNetV2 transfer learning achieves 88.9% accuracy with only ~3,000 images — demonstrating that curated, well-labelled small datasets outperform large but mislabelled ones.
- The RouterBridge RPC model is a powerful abstraction: it cleanly separates real-time MCU tasks from AI/web tasks on the MPU without requiring custom serial protocols.

### 9.3 Future Improvements

- LoRa Mesh Village Network — Deploy one receiver node per village cluster; relay alerts to a central WhatsApp/SMS gateway (currently limited to TX-only with single module).
- Solar Power — TP4056 + 18650 battery + 10W solar panel for fully off-grid autonomous operation.
- Multi-Crop AI Models — Retrain separate Edge Impulse models for cotton, groundnut, and wheat (dominant Gujarat crops); selectable via dashboard.
- Soil NPK Sensor — Add 5-in-1 NPK+EC+pH sensor on UART for comprehensive soil health monitoring.
- Government API Integration — PM-KISAN / Kisan Call Centre API integration for advisory push notifications.
- 4th Irrigation Zone — Add a 4th soil sensor and use Relay Channel 2 for a second pump zone, enabling per-bed precision control.
- Voice Alerts — MPU text-to-speech in Gujarati/Hindi for accessibility to illiterate farmers.

## 10. References

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- Arduino UNO Q Documentation: <https://docs.arduino.cc/hardware/uno-q/>
- Edge Impulse Documentation: <https://docs.edgeimpulse.com/>
- [EcoFarm AI - Deployment - Edge Impulse](#)
- PlantVillage Dataset (Kaggle): <https://www.kaggle.com/datasets/abdallahalidev/plantvillage-dataset>
- Sandeep Mistry LoRa Library: <https://github.com/sandeepmistry/arduino-LoRa>
- U8g2 Library: <https://github.com/olikraus/u8g2>
- GitHub Repository: <https://github.com/HarshParmar029/EcoFarm-AI-UNO-Q>
- Demo Video:  
<https://drive.google.com/file/d/1ffssasdDGP47N2qIOk9EOX2IHYWHtLz/view?usp=sharing>

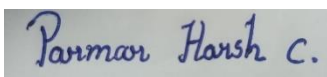
## 11. Declaration

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I, Harsh Chandreshbhai Parmar, student of Marwadi University, Rajkot, Gujarat  
hereby declare that:

- This project — EcoFarm AI — is my own original work, conceived and built independently as a solo participant.
- All hardware components were acquired and tested by me; all code was written, debugged, and verified by me.
- The AI model was trained by me using Edge Impulse Studio with publicly available PlantVillage dataset images.
- No portion of this project has been copied from any other submission or plagiarised from any source without appropriate attribution.
- The claims made in this report (sensor accuracy, water savings, inference time) are based on actual measurements and tests performed on the physical hardware.
- I agree to the Terms and Conditions and Official Rules of the Arduino Physical AI Challenge India 2026.

Signature: \_\_\_\_\_

A rectangular box containing a handwritten signature in blue ink that reads "Parmar Harsh C."

Name: Harsh Chandreshbhai Parmar

Institution: Marwadi University, Rajkot, Gujarat

Date: August 2026